

Provided by: [fig2dev\\_3.2.7a-7ubuntu0.1\\_amd64](#)

## NAME

fig2dev - translates Fig code to various graphics languages

## SYNOPSIS

```
fig2dev -L language [-m mag] [-s fsize] [-Z maxdimension]
        [-D +/-rangelist [-K]] [other options] [fig-file
        [out-file]]
```

## DESCRIPTION

fig2dev translates fig code in the named fig-file into the specified graphics language and puts them in out-file. The default fig-file and out-file are standard input and standard output, respectively

Xfig (Facility for Interactive Generation of figures) is a screen-oriented tool which runs under the X

Window System, and allows the user to draw and manipulate objects interactively. This version of fig2dev

is compatible with xfig versions 1.3, 1.4, 2.0, 2.1, 3.0, 3.1 and 3.2.

Xfig version 3.2.3 and later saves and allows the user to edit comments for each Fig object. These

comments are output with several of the output languages, such as PostScript, CGM, EMF, LaTeX, MetaFont,

PicTeX, (as % comments), tk (as # comments), and pic (as .\" comments).

## GENERAL OPTIONS (all drivers)

-L language

Set the output graphics language. Valid languages are box, cgm, dxf, epic, eepic, eepicemu, emf,

eps, gbx (Gerber beta driver), gif, ibmgl, jpeg, latex, map (HTML image map), mf (MetaFont), mp

(MetaPost), pcx, pdf, pdftex, pdftex\_t, pic, pict2e, pictex, png, ppm, ps, pstex, pstex\_t,

pstricks, ptk (Perl/tk), shape (LaTeX shaped paragraphs), sld (AutoCad slide format), svg, textyl,

tiff, tikz, tk (tcl/tk), tpic, xbm and xpm.

Notes:

You must have ghostscript installed to get the pdf output, and ghostscript and the netpbm package

to get the bitmap formats (png, jpeg, etc.).

-h Print help message with all options for all output languages then exit.

-V Print the program version number and exit.

-D +/-rangelist

With +rangelist, keep only those depths in the list. With -rangelist, keep all depths except those in the list. The rangelist may be a list of comma-separated numbers or ranges separated by colon (:). For example, -D +10,40,55:70,80 means keep only layers 10, 40, 55 through 70, and 80.

-K The selection of the depths with the -D +/-rangelist option does normally not affect the calculation of the bounding box. Thus the generated document might have a much larger bounding box than necessary. If -K is given then the bounding box is adjusted to include only those objects in the selected depths.

-G minor[:major][:unit]

Draws a grid on the page. Specify thin, or thin and thick line spacing in one of several units.

For example, -G .25:1cm draws a thin, gray line every .25 cm and a thicker gray line every 1 cm.

Specifying -G 1in draws a thin line every 1 inch. Fractions may be used, e.g. -G :1/2in will draw a thick line every 1/2 inch.

Allowable units are: i, in, inch, f, ft, feet, c, cm, mm, and m.

Only allowed for PostScript, EPS, PDF, pstricks, tikz and bitmap (GIF, JPEG, etc) drivers.

-j Enable the l18N internationalization facility.

-m mag Set the magnification at which the figure is rendered to mag. The default is 1.0. This may not be used with the maxdimension option (-Z).

-s fsize

Set the default font size (in points, 1/72 inch) for text objects to fsize. The default is

11\*mag, and thus is scaled by the -m option. If there is no scaling, the default font is eleven point Roman.

-Z maxdimension

Scale the figure so that the maximum dimension (width or height) is maxdimension inches or cm, depending on whether the figure was saved with imperial or metric units. This may not be used with the magnification option (-m).

other options

The other options are specific to the choice of graphics language, as described below.

### **OPTIONS COMMON TO ALL BITMAP FORMATS**

-b borderwidth

Make blank border around figure of width borderwidth (1/72 inch).

-F Use correct font sizes (points, 1/72 inch) instead of the traditional size that xfig/fig2dev uses,

which is 1/80 inch. The corresponding xfig command-line option is -correct\_font\_size.

-g color

Use color for the background.

-N Convert all colors to grayscale.

-S smoothfactor

This will smooth the output by passing smoothfactor to ghostscript in the -dTextAlphaBits and -dGraphicsAlphaBits options to improve font rendering and graphic smoothing. A value of 2 for smoothfactor provides some smoothing and 4 provides more.

### **GIF OPTIONS**

-t color

Use color for the transparent color in the GIF file. This must be specified in the same format

that ppmake(1) allows. It may allow an X11 color name, but at least you may use a six-digit

hexadecimal RGBvalue using the # sign, e.g. #ff0000 (Red).

### **JPEG OPTIONS**

-q image\_quality

use the integer value `image_quality` for the JPEG "Quality" factor. Valid values are 0 - 100, with the default being 75.

### **CGM OPTIONS**

CGM is Computer Graphics Metafile, developed by ISO and ANSI and is a vector-based plus bitmap language.

Microsoft WORD, PowerPoint and probably other products can import this format and display it on the screen, something that they won't do with EPS files that have an ASCII preview.

-a Generate binary output.

-r Position arrowheads for CGM viewers that display rounded arrowheads. Normally, arrowheads are pointed, so `fig2dev` compensates for this by moving the endpoint of the line back so the tip of the arrowhead ends where the original endpoint of the line was. If the -r option is used, the position of arrows will NOT be corrected for compensating line width effects, because the rounded arrowhead doesn't extend beyond the endpoint of the line.

### **DXF OPTIONS**

DXF is the Drawing Interchange File Format. The output to DXF is experimental.

-a Select ANSI A paper size instead of the default ISO A4.

-d `xll,yll,xur,yur`

Restrict plotting to a rectangular area of the plotter paper which has a lower left hand corner at (`xll,yll`) and a upper right hand corner at (`xur,yur`). All four numbers are in inches and follow

-d in a comma-separated list - `xll,yll,xur,yur` - with no spaces between them.

-P Rotate the figure to portrait mode. The default is landscape mode.

-v Plot the figure upside-down in portrait mode or backwards in landscape mode.

### **EMF OPTIONS**

EMF is Enhanced Metafile, developed by Microsoft and is a vector-based plus bitmap language. Microsoft

WORD, PowerPoint and probably other products can import this format and display it on the screen,

something that they won't do with EPS files that have an ASCII preview.

-l lang

Set the compatibility level to lang, where lang is one of win95, win98 or winnt. The default is winnt.

-r Position arrowheads for EMF viewers that display rounded arrowheads. See the discussion of the -r option for the CGM output driver above.

## **EPIC OPTIONS**

EPIC is an enhancement to LaTeX picture drawing environment.

EEPIC is an extension to EPIC and LaTeX picture drawing environment which uses tpic specials as a graphics mechanism. It was written by Conrad Kwok of Division of Computer Science at University of California, Davis. Conrad Kwok has also written the EEPIC driver of fig2dev.

EEPIC-EMU is an EEPIC emulation package which does not use tpic specials.

-d factor

Scale arrowheads by factor. The width and height of arrowheads is divided by this factor. This is because EPIC arrowheads are normally about double the size of TeX arrowheads.

-E num Set encoding for text translation (0 = none, 1 = ISO-8859-1, 2 = ISO-8859-2; default 1).

-F Don't set the font face, series, and style; only set it's size and the baselineskip. By default,

fig2dev sets all 5 font parameters when it puts some text. The disadvantage is that you can't set

the font from your LaTeX document. With this option on, you can set the font from your LaTeX document.

If any of the pictures included in your LaTeX document has been generated with -F, then all pictures must be generated with this option.

-f font

Set the default font used for text objects to font, where font is one of rm, bf, it, sf or tt.

The default is rm.

-l lwidth

Use "\thicklines" when the width of the line is equal or wider than lwidth. The default is 2.

-P Generate a complete LaTeX file. In other words, the output file can be formatted without requiring any changes.

-R dummyarg

Allow rotated text. Rotated text will be set using the \rotatebox command. So, you will need to include "\usepackage{graphics}" in the preamble of your LaTeX document. A dummy argument is required after the -R.

If this option is not set, then rotated text will be set horizontally.

-S scale

Set the scale to which the figure is rendered. This option automatically sets the magnification and fsize to scale/12 and scale respectively. Scale must be between 8 and 12, inclusively.

-t stretch

Set the stretch factor of dashed lines to stretch. The default is 30.

-v Include comments in the output file.

-W Enable variable line width. By default, only two line widths are available: The normal line width ("thinlines"), and thick lines ("thicklines"). See also the -l option above.

-w Disable variable line width. Only "\thicklines" and/or "\thinlines" commands will be generated in the output file.

When variable line width option is enabled, the "\thinlines" command is still used when the line width is less than LineThick. One potential problem is that the width of "\thinlines" is 0.4pt but the resolution of Fig is 1/80 inch (approx. 1pt). If LineThick is set to 2, normal lines will be

drawn in 0.4pt wide lines but the next line width is already 2pt. One possible solution is to set LineThick to 1 and set the width of those lines you want to be drawn in "\thinlines" to 0.

Due to this problem, variable line width is disabled by default (-w).

### **IBM-GL (HP/GL) OPTIONS**

IBM-GL (IBM Graphics Language) is compatible with HP-GL (Hewlett-Packard Graphics Language).

-a Select ANSI A paper size instead of the default ISO A4.

-c Generate instructions for an IBM 6180 Color Plotter without an IBM Graphics Enhancement Cartridge (IBM-GEC).

-d xll,yll,xur,yur  
Restrict plotting to a rectangular area of the plotter paper which has a lower left hand corner at (xll,yll) and a upper right hand corner at (xur,yur). All four numbers are in inches and follow  
-d in a comma-separated list - xll,yll,xur,yur - with no spaces between them.

-f fontfile  
Load text character specifications from the table in the file fontfile. The table must have 36 entries - one for each font plus a default. Each entry consists of 5 numbers which specify the  
1.) standard character set (0 - 4, 6 - 9, 30 - 39),  
2.) alternate character set (0 - 4, 6 - 9, 30 - 39),  
3.) character slant angle (degrees),  
4.) character width scale factor and  
5.) character height scale factor.

-k Precede output with PCL command to use HP/GL.

-l pattfile  
Load area fill line patterns from the table in the pattfile file. The table must have 21 entries  
- one for each of the area fill patterns. Each entry consists of 5 numbers which specify the  
1.) pattern number (-1 - 6),  
2.) pattern length (inches),  
3.) fill type (1 - 5),

- 4.) fill spacing (inches) and
- 5.) fill angle (degrees).

-P Rotate the figure to portrait mode. The default is landscape mode.

-p penfile

Load plotter pen specifications from the table in the penfile file. The table must have 9 entries

- one for each color plus a default. Each entry consists of 2 numbers which specify the
  - 1.) pen number (1 - 8) and
  - 2.) pen thickness (millimeters).

-S speed

Set the pen speed to speed (centimeters/second).

-v Plot the figure upside-down in portrait mode or backwards in landscape mode. This allows you to

write on the top surface of overhead transparencies without disturbing the plotter ink on the bottom surface.

-x offset

Shift figure left by offset inches.

-y offset

Shift figure up by offset inches.

Fig2dev may be installed with either ANSI A or ISO A4 default paper size. The -a option selects the

alternate paper size. Fig2dev does not fill closed splines. The IBM-GEC is required to fill other

polygons. Fig2dev may be installed for plotters with or without the IBM-GEC. The -c option selects the

alternate instruction set.

## **LATEX OPTIONS**

-b borderwidth

Make blank border around figure of width borderwidth (1/72 inch).

-d dmag

Set a separate magnification for the length of line dashes to dmag.

-E num Set encoding for latex text translation (0 = no translation, 1 = ISO-8859-1, 2 = ISO-8859-2;

default 1).

-F Don't set the font face, series, and style; only set it's size and the baselineskip. By default, `fig2dev` sets all 5 font parameters when it puts some text. The disadvantage is that you can't set the font from your LaTeX document. With this option on, you can set the font from your LaTeX document.

If any of the pictures included in your LaTeX document has been generated with -F, then all pictures must be generated with this option.

-f font  
Set the default font used for text objects to font, where font is one of rm, bf, it, sf or tt. The default is rm.

-l lwidth  
Sets the threshold between LaTeX thin and thick lines to lwidth pixels. LaTeX supports only two different line widths: `\thinlines` and `\thicklines`. Lines of width greater than lwidth pixels are drawn as `\thicklines`. Also affects the size of dots in dotted line style. The default is 1.

-v Verbose mode. Include comments in the output file.

LaTeX cannot accurately represent all the graphics objects which can be described by Fig. For example, the possible slopes which lines may have are limited. Some objects, such as spline curves, cannot be drawn at all. `Fig2latex` chooses the closest possible line slope, and prints error messages when objects cannot be drawn accurately.

### MAP (HTML image map) OPTIONS

Xfig version 3.2.3 and later saves and allows the user to edit comments for each Fig object. The `fig2dev`

map output language will produce an HTML image map using Fig objects that have `href="some_html_reference"`

in their comments. Any Fig object except compound objects may be used for this. Usually, besides

generating the map file, you would also generate a PNG file, which is the image to which the map refers.

For example, you may have an xfig drawing with an imported image that has the comment `href="go_here.html"`

and a box object with a comment `href="go_away.html"`. This will produce an image map file such the user

may click on the image and the browser will load the "go\_here.html" page, or click on the box and the

browser will load the "go\_away.html" page.

After the map file is generated by `fig2dev` you will need to edit it to fill out any additional

information it may need.

`-b borderwidth`

Make blank border around figure of width `borderwidth` (1/72 inch).

## **METAFONT OPTIONS**

`Fig2dev` scales the figure by 1/8 before generating METAFONT code. The magnification can be further

changed with the `-m` option or by giving magnification options to `mf`.

In order to process the generated METAFONT code, the `mfpic` macros must be installed where `mf` can find

them. The `mfpic` macro package is available at any CTAN cite under the subdirectory: `graphics/mfpic`

`-C code`

Specify the starting METAFONT font code. The default is 32.

`-n name`

Specify the name to use in the output file.

`-p pen_magnification`

Specify how much the line width should be magnified compared to the original figure.

The default

is 1.

`-t top` Specify the top of the whole coordinate system. The default is `ypos`.

`-x xmin`

Specify the minimum x coordinate value of the figure (inches). The default is 0.

`-y ymin`

Specify the minimum y coordinate value of the figure (inches). The default is 0.

-X xmax

Specify the maximum x coordinate value of the figure (inches). The default is 8.

-Y ymax

Specify the maximum y coordinate value of the figure (inches). The default is 8.

## **METAPOST OPTIONS**

-d file

Include file content as additional header.

-i file

Include file content via \input-command.

-M Multipage mode, generate one figure for each depth.

-o Old mode (no latex).

-p number

Adds the line "prologues:=number" to the output.

## **PIC OPTIONS**

-f font

Set the default font used for text objects to font, where font is one of R (roman), B (bold), I (italic), H (sans serif) or C (typewriter). The default is R.

-p ext Enables the use of certain PIC extensions which are known to work with the groff package;

compatibility with DWB PIC is unknown. The extensions enabled by each option are:

arc Allow ARC\_BOX i.e. use rounded corners

line Use the 'line\_thickness' value

fill Allow ellipses to be filled

all Use all of the above

psfont Don't convert PostScript fonts generic type  
(useful for files going to be Dtroff'ed for  
and printed on PS printer). DWB-compatible.

allps Use all of the above (i.e. "all" + "psfont")

## **PICT2E OPTIONS**

PICT2E is an enhancement to the LaTeX picture environment. It is enabled by inserting

`"\usepackage{pict2e}"` in the document preamble. Depending on the content of the figure, it may be necessary to also include `"\usepackage{color}"` and `"\usepackage{graphics}"`. Figures produced with the

PICT2E driver can be processed with any LaTeX engine, e.g., LaTeX + dvips, LaTeX + dvipdfm, pdflatex,

xelatex, ConTeX, etc. Pattern fills are not supported by the PICT2E output language. The PICT2E driver

renders patterns by filling the respective area with the pen-color at 25% intensity, i.e., a 75% tint of

the pen-color. The PICT2E driver allows one to choose any font available to the LaTeX engine, including

PostScript fonts.

`-b borderwidth`

Make blank border around figure of width  $\text{borderwidth} \times (1/72)$  inches.

`-C num` Do not emit a `\color-command` for the color number `num`. (0 = black, 1 = blue, 2 = green - see the

color chooser widget in Xfig). By default, fig2dev does not issue a `\color-command` for objects

which have the color set to "Default" in xfig. With this option, the `"\color"-command` is also

omitted for objects having the color `num`. The color of these objects, as well as of those having

the color set to "Default", is picked up from the including LaTeX-document.

The option `-C 0` is particularly useful. By default, xfig starts with the color set to black.

Then, fig2dev emits `"\color{black}"` commands, and the color-package must be included in the

document preamble. For black text and black-and-white drawings, this is superfluous.

`-e` Do not try to be compatible with epic/epic. By default, you can include `"\usepackage{pict2e,`

`epic, eepic}"` (in this order!) in the document preamble and mix LaTeX pictures using the

epic/epic command set and pictures produced with the PICT2E output language within one document.

With this option on, epic or eepic pictures can not be mixed with PICT2E-pictures.

By default, fig2dev avoids the use of the `"\circle"` and `"\oval"-commands`, which are defined by

epic, in lieu of the `\circlearc`-command exclusive to `pict2e`. In addition, line widths are not only set using `\linethickness`, but also with the `eepic`-command `\allinethickness` (if it is defined).

**-E num** Set encoding for text translation (0 = no translation, 1 = ISO-8859-1, 2 = ISO-8859-2; default 1).

For instance, to use utf8-encoded text, first create a text object, then edit the text using the

edit-button in `xfig`. Convert the `fig`-file to `pict2e` with the option `-E 0` and include `\usepackage[utf8]{inputenc}` in the LaTeX file. In `xfig`, the text typed in may not be displayed

correctly, but the document produced from the LaTeX file will show the same text as was typed in.

**-F** Do not set the font family, series or shape. By default, `fig2dev` sets the font family, series,

shape, font size and `baselineskip`. With this option on, the text font can be set from the

including LaTeX-document, e.g., `\itshape \input{fig1.pict2e}`". See also `-o` (no font size).

**-f font**

Set the default font used for text objects to `font`. The string `font` may be one of `rm`, `bf`, `it`, `sf`,

`tt`, `\rmfamily`, `\bfseries`, `\itshape`, `\sffamily`, `\ttfamily`, or one of the 35 standard PostScript

font names. The default is `\rmfamily`.

**-i dir** Prepend the string `dir` to graphics files included in the `pict2e`-picture. For instance, having

imported `"image.jpg"` in `xfig`, with `-i '$HOME/Figures/'` the code `"\includegraphics{'$HOME/Figures/image.jpg'}`" will be generated.

**-o** Do not set the font size or `baselineskip`. Text will be rendered at the size that is in force where

the `pict2e`-code is inserted into the LaTeX-document, e.g., `\small \input{fig1.pict2e}`". See also

`-F` (no font properties).

**-O** Do not quote characters special to TeX/LaTeX. Useful to get, e.g., an italic `x`, not `$x$`, because

it was forgotten to set the text-flag "special-text" in xfig. This option effectively sets the "special-text" flag for all text.

-P Pagemode, generate a stand-alone LaTeX-file as out-file. The document produced from the LaTeX-file will have the paper size equal to the figure's bounding box (but see the -b option to add a margin). The generated LaTeX-file calls the package "geometry.sty" to set the paper size.

-R num Replace arrowheads num by LaTeX-arrows (" $\vector$ "). The number of an arrowhead ("Arrow Type" in xfig) can be found by opening the arrow chooser widget in xfig and counting the arrows, starting from 1. For instance, to replace filled triangle arrowheads with LaTeX  $\vector$ -commands, use -R 3.

-r Replace all arrows by LaTeX-arrows.

-T Only use TeX fonts, even where PostScript-fonts are specified.

-v Verbose mode. Write comment lines into the output file, usually naming the type of the object that is drawn.

-w Remove the suffix from included graphics-files. With this option on, fig2dev generates code that contains, e.g., " $\includegraphics\{fig1\}$ ", instead of " $\includegraphics\{fig1.eps\}$ ".

## PICTEX OPTIONS

In order to include PiCTeX pictures into a document, it is necessary to load the PiCTeX macros.

PiCTeX uses TeX integer register arithmetic to generate curves, and so it is very slow. PiCTeX draws curves by  $\put$ -ing the psymbol repeatedly, and so requires a large amount of TeX's internal memory, and generates large DVI files. The size of TeX's memory limits the number of plot symbols in a picture. As a result, it is best to use PiCTeX to generate small pictures.

-a Anonymous mode. Do not write the user name into the output file.

-E num Set encoding for latex text translation (0 = no translation, 1 = ISO-8859-1, 2 = ISO-8859-2; default 1).

-f font

Set the default font used for text objects to font, where font is one of rm, bf, it, sf or tt. The default is rm.

-l dimen

Set line thickness to dimen. Default "1pt".

-p psymbol

Set the psymbol. Default "\makebox(0,0)[l]{\tencirc\symbol{'160}}".

-r Do not allow rotated text. Otherwise, files with PiCTeX macros and rotated text need to be processed with dvips.

### **GBX OPTIONS (Gerber, RS-247-X)**

Typically you will wish to set the y scale to -1. See -g for more information.

-d [mm|in]

Output dimensions should be assumed to be millimeters (mm) or inches (in). The default is millimeters.

-p [pos|neg]

Select the image polarity. For positive images lines drawn in the fig file will generate lines of material. For negative images lines drawn in the fig file will result in removed material.

Consider etching a chrome on glass transmission mask. Drawing lines in the fig file and choosing

'neg' will result in these lines being etched through the chrome, leaving transparent lines.

-g <x scale>x<y scale>+<x offset>+<y offset>

This controls the geometry of the output, scaling the dimensions as shown and applying the given

offset. Typically you will wish to set the y scale to -1, mirroring about the x axis. This is

because Gerber assumes the origin to be bottom left, while xfig selects top left.

-f <n digits>.<n digits>

This controls the number of digits of precision before and after the implied decimal point. With

-f 5.3 the following number 12345678 corresponds to 12345.678. Whereas with -f 3.5 it corresponds

to 123.45678. The default is for 3 places before the decimal point and 5 after.

This

corresponds, to a range of 0 to 1m in 10 micron increments.

-v Output comments describing the type of objects being output. The text appears as comments

starting with ## on each line in the output file.

### **POSTSCRIPT, ENCAPSULATED POSTSCRIPT (EPS), and PDF OPTIONS**

With PostScript, xfig can be used to create multiple page figures. Specify the -M option to produce a

multi-page output. For posters, add -O to overlap the pages slightly to get around the problem of the

unprintable area in most printers, then cut and paste the pages together. Great for text with very big letters.

The EPS driver has the following differences from PostScript:

- o No showpage is generated because the output is meant to be imported into another program or

- document and not printed

- o The landscape/portrait options are ignored

- o The centering option is ignored

- o The multiple-page option is ignored

- o The paper size option is ignored

- o The x/y offset options are ignored

The EPS driver has the following two special options:

-B 'Wx [Wy X0 Y0]'

This specifies that the bounding box of the EPS file should have the width Wx and the height Wy.

Note that it doesn't scale the figure to this size, it merely sets the bounding box. If a value

less than or equal to 0 is specified for Wx or Wy, these are set to the width/height respectively

of the figure. Origin is relative to screen (0,0) (upper-left). Wx, Wy, X0 and Y0 are

interpreted in centimeters or inches depending on the measure given in the fig-file.  
Remember to  
put either quotes (") or apostrophes (') to group the arguments to -B.

-R 'Wx [Wy X0 Y0]'

Same as the -B option except that X0 and Y0 is relative to the lower left corner of the figure.

Remember to put either quotes (") or apostrophes (') to group the arguments to -R.

The PDF driver accepts all of the PostScript options, if the -P (pagemode) option is given. In this case, the size of the PDF is the pagesize given in the file or set from the command line via the -z option. Otherwise, if -P is not given, the PDF is cropped to the bounding box of the figure (optionally with a blank border margin set by the -b option), and all of the EPS options are supported.

Text can now include various ISO-character codes above 0x7f, which is useful for language specific characters to be printed directly. Not all ISO-characters are implemented.

Color support: Colored objects created by Fig can be printed on a color postscript printer. There are 32

standard colors: black, yellow, white, gold, five shades of blue, four shades of green, four shades of

cyan, four shades of red, five shades of magenta, four shades of brown, and four shades of pink. In

addition there may be user-defined colors in the file. See the xfig FORMAT3.2 file for the definition of

these colors. On a monochrome printer, colored objects will be mapped into different grayscales by the

printer. Filled objects are printed using the given area fill and color. There are 21 "shades" going

from black to full saturation of the fill color, and 21 more "tints" from full saturation + 1 to white.

In addition, there are 16 patterns such as bricks, diagonal lines, crosshatch, etc.

-A Add an ASCII (EPSI) preview. Not for PDF.

-a Anonymous mode. Do not write the user's login name into the output file.

-b borderwidth

Make blank border around figure of width borderwidth (1/72 inch).

-C dummy\_arg

Add a color \*binary\* TIFF preview for Microsoft products that need a binary preview.

See also -T

(monochrome preview). A dummy argument must be supplied for historical reasons.

Not for PDF

output.

-c Center the figure on the page. The centering may not be accurate if there are texts in the

fig\_file that extends too far to the right of other objects.

-e Put the figure against the edge (not centered) of the page. Not available in EPS.

-F Use correct font sizes (points, 1/72 inch) instead of the traditional size that xfig/fig2dev uses,

which is 1/80 inch. The corresponding xfig command-line option is -correct\_font\_size.

-f font

Set the default font used for text objects to font, where font is one of the 35 standard

PostScript font names. The default is Times-Roman.

-g color

Use color for the background.

-l dummy\_arg

Generate figure in landscape mode. The dummy argument is ignored, but must appear on the command

line for reasons of compatibility. This option will override the orientation specification in the

file (for file versions 3.0 and higher). Not available in EPS.

-M Generate multiple pages if figure exceeds paper size. Not available in EPS.

-N Convert all colors to grayscale.

-n name

Set the Title part of the PostScript output to name. This is useful when the input to fig2dev

comes from standard input.

-O When used with -M, overlaps the pages slightly to get around the problem of the unprintable area

in most printers. Not available in EPS.

-p dummy\_arg

Generate figure in portrait mode. The dummy argument is ignored, but must appear on the command

line for reasons of compatibility. This option will override the orientation specification in the

file (for file versions 3.0 and higher). This is the default for Fig files of version 2.1 or lower. Not available in EPS.

-T Add a monochrome \*binary\* TIFF preview for Microsoft products that need a binary preview. See

also -C (color preview). Not available for PDF output.

-x offset

Shift the figure in the X direction by offset PostScript points (1/72 inch). A negative value

shifts the figure to the left and a positive value to the right. Not available in EPS.

-y offset

Shift the figure in the Y direction by offset points (1/72 inch). A negative value shifts the

figure up and a positive value down. Not available in EPS.

-z papersize

Set the papersize. Not available in EPS.

Available paper sizes are:

Letter (8.5" x 11" also A),

Legal (11" x 14")

Ledger (11" x 17"),

Tabloid (17" x 11", really Ledger in Landscape mode),

A (8.5" x 11" also Letter),

B (11" x 17" also Ledger),

C (17" x 22"),

D (22" x 34"),

E (34" x 44"),

A9 (37 mm x 52 mm),

A8 (52 mm x 74 mm),

A7 (74 mm x 105 mm),

A6 (105 mm x 148 mm),

A5 (148 mm x 210 mm),

A4 (210 mm x 297 mm),

A3 (297 mm x 420 mm),

A2 (420 mm x 594 mm),

A1 (594 mm x 841 mm),

|     |                    |
|-----|--------------------|
| A0  | (841 mm x1189 mm), |
| B10 | ( 32 mm x 45 mm),  |
| B9  | ( 45 mm x 64 mm),  |
| B8  | ( 64 mm x 91 mm),  |
| B7  | ( 91 mm x 128 mm), |
| B6  | (128 mm x 182 mm), |
| B5  | (182 mm x 257 mm), |
| B4  | (257 mm x 364 mm), |
| B3  | (364 mm x 515 mm), |
| B2  | (515 mm x 728 mm), |
| B1  | (728 mm x1030 mm), |
| B0  | (1030mm x1456 mm). |

### **PSTEX and PDFTEX OPTIONS**

The pstex and pdftex languages are a variant of ps which suppress text that has the text flag "TeX Text"

set. The pstex\_t and pdftex\_t languages have the complementary behavior: they generate only the text

that has the "Tex Text" flag set and the commands necessary to position this text. They also generate the

commands necessary to overlay the PostScript or PDF file generated using pstex/pdftex.

These two drivers

can be used to generate a figure which combines the flexibility of PostScript graphics with LaTeX text

formatting of text flagged as "TeX Text".

The pstex and pdftex drivers accept the same options that the EPS driver accepts.

-n name

Set the Title part of the PostScript output to name. This is useful when the input to fig2dev

comes from standard input.

### **PSTEX\_T and PDFTEX\_T OPTIONS**

The pstex\_t and pdftex\_t languages produce only the text flagged with the "TeX Text" flag, the commands

necessary to position this text, and the commands necessary to overlay the PostScript or PDF file

generated using pstex or pdftex (see above).

-E num Set encoding for latex text translation (0 no translation, 1 ISO-8859-1, 2 ISO-8859-2; default 1)

-F Don't set the font face, series, and style; only set it's size and the baselineskip. By default, fig2dev sets all 5 font parameters when it puts some text. The disadvantage is that you can't set the font from your LaTeX document. With this option on, you can set the font from your LaTeX document (like "\sfshape \input picture.eepic").

-p file specifies the name of the PostScript file to be overlaid. If not set or its value is null then no PS file will be inserted.

### **PSTricks OPTIONS**

The PSTricks driver provides full LaTeX text and math formatting for XFig drawings without overlaying separate outputs as in the PSTEX methods. The output matches the quality of output of the PostScript driver except for text, where the Latex font selection mechanism is used as for other fig2dev LaTeX drivers. In addition, text is rendered black, although font color-changing LaTeX code can be embedded in the drawing. The generated PSTricks code is meant to be readable. Each command stands alone, not relying on global option state variables. Thus the user can easily use XFig to rough out a PSTricks drawing, then finish by hand editing.

To use the driver's output, give the command "\usepackage{pstricks}" in your document preamble. The graphicx and pstricks-add packages may also be required. The former is used for bitmap graphics and the second for complex line styles and/or hollow PSTricks arrows (with the -R 1 option). The driver will tell you which packages are needed. In the document body, include the figure with "\input{pstfile}" where pstfile.tex is the output file. Use the XFig special flag to have text passed as-is to LaTeX. For non-special text, the same mechanism as the LaTeX and epic driver mechanism is used to match font specs, but this is imprecise.

Known bugs and limitations.

PSTricks support for join styles is version dependent. Raw postscript is inserted with `"\pstVerb"`

for old versions when other than angle joins are needed. The `-t` option controls this behavior.

PSTricks does not support rotated ellipses directly, so a `rput` command is emitted that rotates and

locates a horizontal ellipse. This makes a problem with hatch patterns, which are moved and

rotated along with the ellipse. Hatch rotation is fixed by a counter-rotation, but the origin is

not adjusted, so registration with adjacent hatch patterns will be incorrect. Flipped bitmap

graphics use an undocumented feature of the `graphicx` package: a negative height flips the image

vertically. This appears to work reliably. However, you may want to flip graphics with another

program before including them in Xfig drawings just to be sure. With the `-p` option, the driver

attempts to convert non-EPS pictures to EPS with the TeX distribution's `bmeps` program, but `bmeps`

does not know about very many file formats including gif.

`-f font`

Set the default font used for text objects to `font`, where `font` is one of `rm`, `bf`, `it`, `sf` or `tt`. The default is `rm`.

`-G dummy_arg`

Draws a standard PSTricks grid in light gray, ignoring the size parameters, numbered in PSTricks units.

`-l weight`

Sets a line weight factor that is multiplied by the actual Fig line width. The default value 0.5 roughly matches the output of the PS driver.

`-n 0|1|2|3`

Sets environment type. Default 0 creates a `\picture` environment with bounding box exactly

enclosing the picture (but see `-x` and `-y`). A 1 emits bare PSTricks commands with no environment

at all, which can be used with `\input{commands}` inside an existing `\pspicture`. A 2 emits a

complete LaTeX document. A 3 also emits a complete LaTeX document but attempts to set the PSTricks unit to fit a 7.5 by 10 inch (portrait aspect) box.

-P Shorthand for -n 3.

-p dir Attempts to run the bmeps program to translate picture files to EPS, which is required by

PSTricks. The translated files go in dir , which must already exist (the driver will not create

it). Moreover, (BIG CAVEAT HERE) the driver overwrites files with impunity in this directory!

Don't put your stuff here. The includegraphics commands in the output file refer to this

directory. Even if the -p option is not used, includegraphics commands follow this convention

with the default directory ./eps . In this case, the user must do the conversions independently.

The bmeps program is part of the standard TeX distribution. It converts the following formats to

EPS: png jpg pnm tif. You can see the bmeps command with the -v option.

-R 0|1|2

Sets arrow style. With the default style 0, Fig arrows are converted to lines and polygons. With

style 1, the Fig arrowhead dimensions are converted to PSTricks arrowhead dimensions and PSTricks

arrowhead options are emitted. Hollow arrows will require the additional package pstricks-add.

With style 2, PSTricks arrowhead options are emitted with no dimensions at all, and arrowhead size

may be controlled globally with psset.

-S scale

Scales the image according to the same convention as the EPIC driver, i.e., to size scale/12.

-t version

Provides the driver with PSTricks version number so output can match expected LaTeX input.

-v Print verbose warnings and extra comments in the output file. Information provided includes font

substitution details, the bmeps commands used for picture conversion, if any, and one comment per Fig object in the output.

-x marginsize

Adds marginsize on the left and right of the PStricks bounding box. By default, the box exactly encloses the image.

-y marginsize

Adds marginsize on the top and bottom of the PStricks bounding box. By default, the box exactly encloses the image.

-z 0|1|2

Sets font handling option. Default option 0 attempts to honor Fig font names and sizes, finding the best match with a standard LaTeX font. Option 1 sets LaTeX font size only. Option 2 issues no font commands at all.

## TEXTYL OPTIONS

-f font

Set the default font used for text objects to font, where font is one of rm, bf, it, sf or tt. The default is rm.

-l lwidth

Set the line thickness. lwidth must be a value between 1 and 12.

## TIKZ OPTIONS

TIKZ is a powerful frontend to the Portable Graphics Format (PGF) developed by Till Tantau, now at the

University of Lübeck. TIKZ was developed to be as platform-independent as possible, i.e., tikz-code can

be processed with plain TeX, pdftex, xetex, LaTeX, ConTeX, pdfflatex, lualatex, or combinations of LaTeX +

dvips, LaTeX + dvipdfm or others. The TIKZ-code emitted by fig2dev tries to maintain this portability.

For instance, a tikz-picture is commenced with \tikzpicture (TeX-style), to not exclude any processing

engine. However, the stand-alone file produced with the -P option must be processed with a LaTeX-engine.

In addition, font-commands may require a LaTeX engine.

-b borderwidth

Make blank border around figure of width borderwidth\*(1/72) inches.

-C num Do not emit a \color-command for the color number num. (0 = black, 1 = blue, 2 = green - see the color chooser widget in Xfig). By default, fig2dev does not issue a \color-command for objects

which have the color set to "Default" in xfig. With this option, the "\color"-command is also

omitted for objects having the color num. The color of these objects, as well as of those having

the color set to "Default", is picked up from the including document.

-E num Set encoding for text translation (0 = no translation, 1 = ISO-8859-1, 2 = ISO-8859-2; default 1).

For instance, to use utf8-encoded text, first create a text object, then edit the text using the

edit-button in xfig. Convert the fig-file to tikz with the option -E 0 and include "\usepackage[utf8]{inputenc}" in the LaTeX file (not necessary when using xelatex). In xfig, the

text typed in may not be displayed correctly, but the document produced from the LaTeX file will

show the same text as was typed in.

-F Do not set the font family, series or shape. By default, fig2dev sets the font family, series,

shape, font size and baselineskip. As a side effect, this requires the New Font Selection Scheme

(NFSS) of LaTeX. With this option on, the text font can be set from the including document, which

may be TeX or LaTeX. See also -o (no font size).

-f font

Set the default font used for text objects to font. The string font may be one of rm, bf, it, sf,

tt, \rmfamily, \bfseries, \itshape, \sffamily, \ttfamily, or one of the 35 standard PostScript

font names. The default is \rmfamily.

-i dir Prepend the string dir to graphics files included in the tikz-picture. For instance, having

imported "image.jpg" in xfig, with - i '\$HOME/Figures/' the code "\pgfimage[width=..., height=...]{ \$HOME/Figures/image.jpg}" will be generated.

-O Do not quote characters special to TeX/LaTeX. Useful to get, e.g., an italic x, not  $x$ , because

it was forgotten to set the text-flag "special-text" in xfig. This option effectively sets the

"special-text" flag for all text.

-o Do not set the font size or baselineskip. Text will be rendered at the size that is in force where

the tikz-code is inserted into the document, e.g., `"\small\input fig1.tikz"`. See also -F (no font

properties).

-P Pagemode, generate a stand-alone LaTeX-file as out-file. The document produced from the LaTeX-file

will have the paper size equal to the figure's bounding box (but see the -b option to add a

margin). The generated LaTeX-file calls the package "geometry.sty" to set the paper size.

-T Only use TeX fonts, even where PostScript-fonts are specified.

-v Verbose mode. Write comment lines into the output file, usually naming the type of the object that

is drawn.

-W Do not emit code at the beginning of the file that allows one to set the figure width or height

from the including TeX document. Otherwise, e.g.,

`"\newdimen\XFigwidth\XFigwidth=\linewidth"`

would scale the following figures to the line width.

-w Remove the suffix from included graphics-files. With this option on, fig2dev generates code that

contains, e.g., `"\pgfimage{fig1}"` instead of `"\pgfimage{fig1.pdf}"`.

### **TK and PTK OPTIONS (tcl/tk and Perl/tk)**

Arc-boxes are not supported for the tk output language, and only X bitmap pictures are supported because

of the canvas limitation in tk.

Picture objects are not scaled with the magnification factor for tk output.

Because tk scales canvas items according to the X display resolution, polygons, lines, etc. may be scaled

differently than imported pictures (bitmaps) which aren't scaled at all.

-g color

Use color for the background.

-l dummy\_arg

Generate figure in landscape mode. The dummy argument is ignored, but must appear on the command

line for reasons of compatibility. This option will override the orientation specification in the

file (for file versions 3.0 and higher).

-p dummy\_arg

Generate figure in portrait mode. The dummy argument is ignored, but must appear on the command

line for reasons of compatibility. This option will override the orientation specification in the

file (for file versions 3.0 and higher). This is the default for Fig files of version 2.1 or lower.

-P Generate canvas of full page size instead of using the bounding box of the figure's objects. The

default is to use only the bounding box.

-w Wrap the figure with code in order to generate a complete perl file. That is, you can do fig2dev

-L ptk -w f.fig f.pl; perl f.pl and a widget pops up that shows the graphics contained in f.pl.

Only available for ptk output.

-z papersize

Set the paper size. See the POSTSCRIPT OPTIONS for available paper sizes. This is only used when

the -P option (use full page) is used.

## TPIC OPTIONS

-f font

Set the default font used for text objects to font. The default is rm. The string font can be

one of rm, bf, it, sf, tt, avant, avantcsc, avantd, avantdi, avanti, bookd, bookdi, bookl,

booklcsc, bookli, chanc, cour, courb, courbi, couri, helv, helvb, helvbi, helvc, helvcb, helvcbi,

helvci, helvcsc, helvi, pal, palb, palbi, palbu, palc, palcsc, pali, palsl, palu, palx, times,

timesb, timesbi, timesc, timescsc, timesi, timesl or timesx.

## **SEE ALSO**

xfig(1), pic(1), pic2fig(1), transfig(1)

## **BUGS and RESTRICTIONS**

Please send bug reports, fixes, new features etc. to:  
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